

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 1

Algebra, Quadratics & Surds

Duration: 2.5 hours

Total Marks: 100

Name: _____

Date: _____

SECTION A: Short Answer Questions (40 marks)

1. Solve the quadratic equation: $2x^2 - 5x + 2 = 0$ [4 marks]

2. Simplify: $\sqrt{12} + \sqrt{27} - \sqrt{3}$ [3 marks]

3. If α and β are roots of $x^2 - 3x + 1 = 0$, find $\alpha + \beta$ and $\alpha\beta$ [3 marks]

4. Factorize: $6x^2 + 11x - 10$ [3 marks]

5. Solve: $2^x = 8$ [2 marks]

6. Expand: $(2x + 3)^3$ [3 marks]

7. Solve the inequality: $-2 < 3x - 1 \leq 5$ [3 marks]

8. If $\log_2(x) = 3$, find x [2 marks]

SECTION B: Long Answer Questions (60 marks)

9. Solve simultaneously: $x + y = 5$ and $x^2 + y^2 = 13$ (6 marks)

10. A quadratic has roots -2 and 3. Write down the equation. (4 marks)

11. Solve: $|x - 3| = 2x - 1$ (5 marks)

12. If $f(x) = x^2 - 4x + 3$, find: (a) $f(0)$, (b) values of x where $f(x) = 0$, (c) the minimum value (6 marks)

13. Simplify: $(8x^3)^{2/3} \times x^{-1}$ (5 marks)

14. Prove that $(a - b)^2 + (b - c)^2 + (c - a)^2 = 2(a^2 + b^2 + c^2 - ab - bc - ca)$ (4 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Solve $2x^2 - 5x + 2 = 0$

Using quadratic formula: $x = (5 \pm \sqrt{(25-16)})/4 = (5 \pm 3)/4$, $x = 2$ or $x = 0.5$

Q2: Simplify $\sqrt{12} + \sqrt{27} - \sqrt{3}$

$$= 2\sqrt{3} + 3\sqrt{3} - \sqrt{3} = 4\sqrt{3}$$

Q3: Sum and product of roots

For $x^2 - 3x + 1 = 0$: $\alpha + \beta = 3$, $\alpha\beta = 1$

Q4: Factorize $6x^2 + 11x - 10$

$$(2x + 5)(3x - 2)$$

Q5: Solve $2^x = 8$

$$2^x = 2^3, x = 3$$

Q6: Expand $(2x + 3)^3$

$$= 8x^3 + 36x^2 + 54x + 27$$

Q7: Solve $-2 < 3x - 1 \leq 5$

$$-1/3 < x \leq 2$$

Q8: Find x if $\log_2(x) = 3$

$$x = 2^3 = 8$$

SECTION B SOLUTIONS

Q9: Solve $x + y = 5$ and $x^2 + y^2 = 13$

From first: $y = 5 - x$. Substitute: $x^2 + (5-x)^2 = 13$, $x^2 + 25 - 10x + x^2 = 13$, $2x^2 - 10x + 12 = 0$, $x^2 - 5x + 6 = 0$, $(x-2)(x-3) = 0$, $x = 2$ or 3 . If $x = 2$, $y = 3$; if $x = 3$, $y = 2$. Solutions: $(2,3)$ and $(3,2)$

Q10: Quadratic with roots -2 and 3

$$(x + 2)(x - 3) = 0, x^2 - x - 6 = 0$$

Q11: Solve $|x - 3| = 2x - 1$

Case 1: $x - 3 = 2x - 1$, $-3 + 1 = x$, $x = -2$. Check: $|-2-3| = 5$, $2(-2)-1 = -5$. No. Case 2: $-(x-3) = 2x - 1$, $-x + 3 = 2x - 1$, $4 = 3x$, $x = 4/3$. Check: $|4/3 - 3| = 5/3$, $2(4/3) - 1 = 5/3$. Yes. $x = 4/3$

Q12: $f(x) = x^2 - 4x + 3$

(a) $f(0) = 3$, (b) $x^2 - 4x + 3 = 0$, $(x-1)(x-3) = 0$, $x = 1$ or 3 , (c) Min value at $x = 2$: $f(2) = 4 - 8 + 3 = -1$

Q13: Simplify $(8x^3)^{2/3} \times x^{-1}$

$= (8^{2/3})(x^3)^{2/3} \times x^{-1} = 4x^2 \times x^{-1} = 4x$

Q14: Prove identity

$\text{LHS} = a^2 - 2ab + b^2 + b^2 - 2bc + c^2 + c^2 - 2ca + a^2 = 2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = 2(a^2 + b^2 + c^2 - ab - bc - ca) = \text{RHS}$